

Project Smart Trigger for ECE/CS 3992

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Abstract—Some people will read this and nothing else. What should they know about your proposal?

Index Terms—IEEE, IEEEtran, journal, L^AT_EX, paper, template.

I. PROBLEM STATEMENT

What is the problem the project is trying to solve?

II. SOLUTION SPACE

Efforts to improve firearm safety through access control have evolved through several distinct technological approaches. At the most basic level, traditional solutions rely on purely mechanical safeguards such as trigger locks or integrated key-based locking mechanisms. These systems are inexpensive and highly reliable because they do not depend on electronics or power, but they fundamentally lack user-specific authentication. Anyone with access to the key or combination can operate the firearm, which causes concern toward illegitimate use. In time-sensitive or high-stress situations, the need to manually unlock a device with a key can render the safety ineffective in practice, leading many police and military personnel to avoid using such systems altogether. This creates a gap between theoretical safety and real-world adoption, motivating the development of systems that can enforce access control without requiring manual, non-instantaneous intervention.

Electronic authentication methods represent the next step in this progression. Systems based on PIN entry, external key fobs, or smartphone connectivity attempt to introduce user-specific control while maintaining relatively simple hardware implementations. While these approaches improve upon mechanical systems by enabling programmable access, they still require explicit interaction from the user, such as entering a code or activating a device. This introduces latency and cognitive load that can be unacceptable in defensive scenarios. Additionally, wireless-based systems may be vulnerable to interception or spoofing, depending on their implementation. As a result, these approaches highlight the need for authentication mechanisms that are both secure

and implicit, allowing the system to verify the user seamlessly without requiring conscious action.

Biometric authentication has emerged as the most prominent alternative in this space, particularly in commercial “smart gun” designs. By leveraging physiological traits such as fingerprints, palm vein patterns, or even grip characteristics, biometric systems aim to bind firearm operation directly to the identity of the user. This approach is compelling because it eliminates the need for external tokens and ensures that only authorized individuals can operate the weapon. However, when examined in the context of real-world firearm use, biometrics introduce a number of critical challenges. Sensors such as fingerprint scanners are highly sensitive to environmental conditions. Metrics like dirt, sweat, gloves, or moisture can significantly degrade performance. Firearms are often used in unpredictable and harsh environments, where such conditions are common, leading to an increased likelihood of authentication failure.

Latency is another major concern. Biometric systems require time to capture, process, and match data against stored templates. Even small delays can be unacceptable in scenarios where immediate response is necessary. Equally important is the issue of reliability in terms of false negatives and false positives. A false negative, where an authorized user is denied access, represents a potentially dangerous failure in a defensive situation. Conversely, a false positive undermines the entire purpose of the system. These risks are compounded by the long-term degradation of sensors and the variability of human biological signals. In addition to these technical concerns, biometric systems tend to be significantly more expensive and power-intensive than alternative solutions. Commercial implementations often cost on the order of \$1000 to \$1500, reflecting the complexity of most biometric devices. This cost and power burden conflicts directly with the constraints of a small, battery-powered module designed to fit within the firing control group of an AR-15.

RFID-based authentication offers a different point in the design space by prioritizing simplicity, robustness, and low power consumption. In a passive near-field

RFID system, authentication occurs when a tag enters the electromagnetic field generated by a reader, allowing the tag to transmit its unique identifier without requiring its own power source. This enables extremely low-latency operation and reduces the system's overall energy requirements. RFID systems are also less sensitive to environmental contaminants compared to optical or capacitive biometric sensors, making them more reliable in varied conditions. The primary tradeoff is that RFID relies on possession rather than inherent identity. A user must carry an external token, such as a ring or card, which introduces the possibility of loss or unauthorized use if the token is compromised.

More complex hybrid systems attempt to combine multiple authentication methods, such as pairing RFID with biometrics or behavioral sensing, in order to improve security. While these approaches can reduce certain vulnerabilities, they significantly increase system complexity, cost, and power consumption. For a compact, retrofittable module operating under strict physical and electrical constraints, this hybrid complexity is impractical. As a result, the solution space reflects a series of tradeoffs between security, reliability, usability, and feasibility. Biometric systems offer strong identity assurance but struggle with robustness and cost, while RFID systems sacrifice some degree of identity binding in exchange for speed, reliability, and practicality. This balance makes RFID a compelling candidate for a system intended to be compact, cost-effective, and deployable in real-world conditions.

III. RELATED WORK

A more in-depth, formal exploration of the research and product development related to the problem space.

IV. SOLUTION / CONTRIBUTION

1-2 paragraphs summarizing your approach, what's exciting about it, why it will succeed

V. SAFETY, GLOBAL, SOCIETAL, ETHICAL CONSIDERATIONS

What are the public health, safety and welfare, global, cultural, societal, environmental, and economic factors related to the problem and solution space? What power does the proposed system provide, and to whom? What are the relevant professional ethical obligations to consider in the design of the system?

A. Factors Related to the Problem and Solution

1) Public Health:

2) Safety and Welfare:

3) Global:

4) Cultural:

5) Societal:

6) Environmental:

7) Economic:

B. Power

VI. PROPOSED WORK

A project includes some things and excludes others you know you won't work on. Define more explicitly what is included and not included here. What will be demonstrated at project completion at Demo Day?

VII. PROJECT TASKS

Describe each task.

- "baseline deliverables" – things you will deliver
- "stretch goals" – things that you are planning to include, but the project will succeed even without them (and why)
- What components will be purchased and what will be built?
- What resources are needed to complete the tasks? These may be your own, or in a lab, or may be people who will help you.
- Task Interfaces: What are the important inputs and outputs of each individual component? Define them.
- Testing: How will each task be tested?
- Design approach: For each component, how will it be designed? What tools will be used (software, hardware, measurement, etc.)
- Current status: What tasks have been finished? What is left to be done?
- Schedule: A date when each task will be completed.
- Parts: Complete parts list with vendor name and information

• Pre-Planning:

• Functional RFID reader:

- software for programming outside of ide
- Successful blocking of fire control
- part fitting
- assembly
- endurance testing

VIII. INTEGRATION STRATEGY

How will the individual tasks/components be integrated? How will the integration be tested?

IX. RISK ASSESSMENT

To ensure project success and the safety of all personnel, we have identified and mitigated key risks associated with the development of a smart trigger. Each risk was initially categorized as Red (high), Yellow (moderate), or Green (low). Through a proactive de-risking process, all identified hazards have been moved to the Green category.

- **Safety/Legal (Initially Red):** Bringing a firearm to campus; potential regulatory non-compliance; negligent discharges.
- **Operational (Initially Yellow):** Inadequate real-world testing; mechanical component failure; pinch injuries from moving parts.
- **Design (Initially Green):** Spatial constraints within the fire control group.

Campus Safety and Legal Compliance: The most critical risk involved the presence of a firearm on school grounds. We have fully de-risked this by utilizing a modular design: only the smart trigger assembly is brought to campus for testing, while the firearm remains off-site. To address legal concerns, we are consulting with university police and safety officers to ensure all protocols meet local regulations. These measures transition these risks to **Green**.

Safe Handling and Negligent Discharge: To mitigate risks of improper handling or accidental discharge, all team members have undergone a safety orientation focused on the four primary laws of firearm safety. Furthermore, all initial functionality testing of the smart trigger is performed in a lab environment without a firearm present. These strict protocols ensure a **Green** safety rating.

Testing and Mechanical Reliability: To address the risk of component failure or inadequate testing, we will implement a rigorous, multi-stage validation process. This includes individual component stress testing (e.g., solenoid durability) followed by integrated field testing in varied environments. This ensures the system is robust and functions as intended, moving these concerns to **Green**.

Physical Hazards and Spatial Design: Potential for minor injuries from moving parts is mitigated through the mandatory use of personal protective equipment (PPE) and ergonomic housing designs. Regarding spatial constraints, we will utilize precise measurements of the fire control group and detailed prototypes during the design phase to ensure a proper fit. Both risks are confirmed **Green**.

With these mitigation strategies finalized, the

project is fully de-risked and prepared for safe demonstration.

X. CONCLUSION

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